

LAB 1: AWS Glue Lab: Data Crawling & Cataloging

LAB OVERVIEW

Demonstrate how to create an AWS Glue database and crawler with two data sources using the AWS Glue console.

STEP-BY-STEP PROCEDURES

STEP 1: Access AWS Glue Service

- Search for "AWS Glue" in the AWS console search box
- Choose AWS Glue from Services
- Expand left navigation pane (hamburger menu)
- Navigate to: Data Catalog → Crawlers
- Click "Create crawler"

STEP 2: Set Crawler Properties

- **Crawler Name:** `learn-glue-crawler`
- Click Next

STEP 3: Choose Data Sources & Classifiers

- Data source configuration: Choose "Not yet"
- Click "Add a data source"
- **S3 Path:** Browse S3 and select the bucket starting with `getting-started-with-glue-studio` (note: your bucket has a unique string at the end)
- Select the **datafiles folder**
- Subsequent crawler runs: Choose "Crawl all subfolders"
- Click "Add an S3 data source"
- Verify S3 data source and click Next

STEP 4: Configure Security Settings

- Click "Create new IAM role"
- In the pop-up, append `learn-glue-role` to the existing text
- **Full IAM Role Name:** `AWSGlueServiceRole-learn-glue-role`
- Click Create
- Confirm green banner appears indicating role creation
- Click Next

STEP 5: Set Output & Scheduling

- Click "Add database" (opens in new tab)
- **Database Name:** `sales_data`
- Click "Create database"
- Return to Set output and scheduling page
- Refresh Target database list (circular arrow icon)
- **Target database:** Choose `sales_data`
- **Frequency:** Keep "On demand" selected
- Scroll down and Click "Next"

STEP 6: Review & Create

- Click "Create crawler"

STEP 7: Run the Crawler

- On the Left pane click on Crawler
- Select your Crawler
- Click "Run"
- Monitor status in Crawler runs section:
 - Initial status: Running
 - Completion status: Ready
 - Result: Two tables added to `sales_data` database

STEP 8: Verify Results

- Navigate to Data Catalog → Databases → Tables
 - Confirm two tables created:
 - **sales table**
 - **customers table**
 - These tables are based on analyzed data files in Amazon S3
 - Click on the sales table to browse metadata
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KEY DELIVERABLES

- ✓ Crawler created and named: `learn-glue-crawler`
- ✓ Database created and named: `sales_data`
- ✓ Two tables created: `sales` and `customers`
- ✓ IAM role created: `AWSGlueServiceRole-learn-glue-role`
- ✓ Data source: S3 datafiles folder with all subfolders crawled
- ✓ Metadata populated in Data Catalog